

strategy. If markets are undervalued, the traditional type of declining equity glide path in retirement can look more attractive, as it provides a closer approximation to what would be done with a valuation-based strategy.

In a highly valued market environment (like 2017), retirees can reduce their vulnerability to the effects of a big drop in the stock market by using a lower equity allocation. This can be accomplished by using either a preset rising equity glide path (as an inverse of what today's target-date funds do for the preretirement period), a valuation-based strategy, or some combination of the two.

It must be noted that implementing a valuation-based allocation strategy requires a certain disposition, as it is a contrarian strategy requiring lower stock allocations when people are most giddy about stocks, and requiring higher stock allocations when panic has set in for the typical investor. The fact that the strategies "worked" historically does not guarantee future success. Nonetheless, this research proposes a potential asset allocation approach for retirees wishing to incorporate valuations into their asset allocation choices, but also wishing to maintain a formal commitment to an asset allocation decision framework that will hopefully help prevent hasty, emotion-based decisions. At the very least, these findings can help support the psychological discipline to stay the course and not buy high and sell low.

3. Funded Ratio

Finally, asset allocation can be based on a retiree's funded ratio. The point of calculating an individual's funded ratio is to treat personal retirement planning in the same manner as a corporate pension fund. Are the lifetime values of assets large enough to meet the lifetime liabilities? The answer to this question is measured through the funded ratio. A funded status of 100 percent means that a household has just enough assets to meet liabilities. Overfunded and underfunded individuals have more or less than this.

This approach seeks to reduce portfolio volatility when retirees have enough assets to just get by with meeting their retirement spending goals using a low-volatility portfolio. Once retirees have excess "discretionary" wealth beyond what is needed to safely lock in their goals, they can invest more aggressively with a volatile portfolio. In other words, reduce volatility when the possibility